

# COMPLEX ANALYSIS GENERAL EXAM JANUARY 2025

Solve as many problems as you can. Full solutions on a smaller number of problems will be worth more than partial solutions on several problems. Don't use any of the Picard theorems. Throughout,  $\mathbb{D}$  denotes the unit disc  $\{z \in \mathbb{C} : |z| < 1\}$ .

## Problem 1

Let  $f : \overline{\mathbb{D}} \rightarrow \mathbb{C}$  be a continuous function for which  $f|_{\mathbb{D}}$  is holomorphic and  $|f(z)| = 1$  for all  $z \in \partial\mathbb{D}$ . Show that there exists a finite collection of points  $z_1, \dots, z_n \in \mathbb{D}$  and some  $\theta \in \mathbb{R}$  such that

$$f(z) = e^{i\theta} \prod_{j=1}^n \frac{z_j - z}{1 - \overline{z_j}z}$$

## Problem 2

Let  $f : \mathbb{C} \setminus \{2, 3\} \rightarrow \mathbb{C}$  be the holomorphic function

$$f(z) := \frac{z}{(z-2)(z-3)}$$

For each of the following regions, determine the Laurent series of  $f$  that converges on the given region.

- (a)  $\{z \in \mathbb{C} : |z| < 2\}$
- (b)  $\{z \in \mathbb{C} : 2 < |z| < 3\}$
- (c)  $\{z \in \mathbb{C} : |z| > 3\}$

## Problem 3

Let  $u : \mathbb{R}^2 \rightarrow \mathbb{R}$  be a harmonic function, and suppose that  $u(x, y) \leq x$  for all  $(x, y) \in \mathbb{R}^2$ . Show that there exists a constant  $c \in (-\infty, 0]$  such that  $u(x, y) = x + c$  for all  $(x, y) \in \mathbb{R}^2$ .

## Problem 4

Using Cauchy's residue theorem, show that for  $a, b \in \mathbb{R}$  with  $0 < b < a$ ,

$$\int_0^{2\pi} \frac{1}{(a + b \sin \theta)^2} d\theta = \frac{2\pi a}{(a^2 - b^2)^{3/2}}$$

## ***Problem 5***

- (a) Let  $f : \mathbb{D} \rightarrow \mathbb{D}$  be a holomorphic function satisfying  $f(0) = \frac{2}{3}$ . Show that  $|f'(0)| \leq \frac{5}{9}$ .
- (b) Determine all holomorphic functions  $f : \mathbb{D} \rightarrow \mathbb{D}$  that simultaneously satisfy both  $f(0) = \frac{2}{3}$  and  $|f'(0)| = \frac{5}{9}$ .